SMOOTH BASE CHANGE FOR A LIMIT OF SMOOTH SCHEMES AND TORSION COEFFICIENTS INVERTIBLE ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the two spectral sequences, the identification of their entries from the two base change inputs, and the resulting identification of the abutments.

Editorial note. The label lemma-smooth-base-change-general names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement's fibre square is the author's and is delivered verbatim; the three cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- The smooth base change theorem. Same chapter, the smooth base change theorem for a smooth morphism with torsion coefficients invertible on the base. Supplies the identification g^{-1}R^p f_* of a single sheaf with the derived pushforward along f'.
- Base change along a limit of smooth schemes. Same chapter, the base change lemma for a directed inverse limit of schemes smooth over the base. Supplies the extension of that identification to the limit situation.
- The spectral sequence of a complex computing the pushforward. Derived Categories, the lemma giving the two spectral sequences E_2^{p,q} = R^p f_*H^q(E) and the corresponding one for f' converging to the derived pushforwards. Supplies the comparison of the two spectral sequences that yields the statement for complexes.

Source reference. Original lemma 015J: sources/stacks-chow/tags.txt, source lines 1510–1510. Prerequisite pointer invoked at proof line(s) 18003; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EZT: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 16410–16410. Prerequisite pointer invoked at proof line(s) 18006; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 0EYU: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 17446–17446. Prerequisite pointer invoked at proof line(s) 18005; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $S' = \lim S_i$ be a directed inverse limit of schemes S_i smooth over S with affine transition morphisms. Let $f : X \to S$ be quasi-compact and quasi-separated and form the fibre square*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$

Then

$$g^{-1}Rf_*E = R(f')_*(g')^{-1}E$$

for any $E \in D^+(X_{étale})$ whose cohomology sheaves $H^q(E)$ have stalks which are torsion of orders invertible on S.

Proof. Consider the spectral sequences

$$E_2^{p,q} = R^pf_*H^q(E) \quad \text{and} \quad {E'}_2^{p,q} = R^pf'_*H^q((g')^{-1}E) = R^pf'_*(g')^{-1}H^q(E)$$

converging to R^nf_*E and $R^nf'_*(g')^{-1}E$. These spectral sequences are constructed in Derived Categories, Lemma 015J. Combining the smooth base change theorem (Theorem 0EYU) with Lemma 0EZT we see that

$$g^{-1}R^pf_*H^q(E) = R^p(f')_*(g')^{-1}H^q(E)$$

Combining all of the above we get the lemma. □